國立臺北科技大學
National Taipei University of Technology

資料結構
Data Structure

加分題

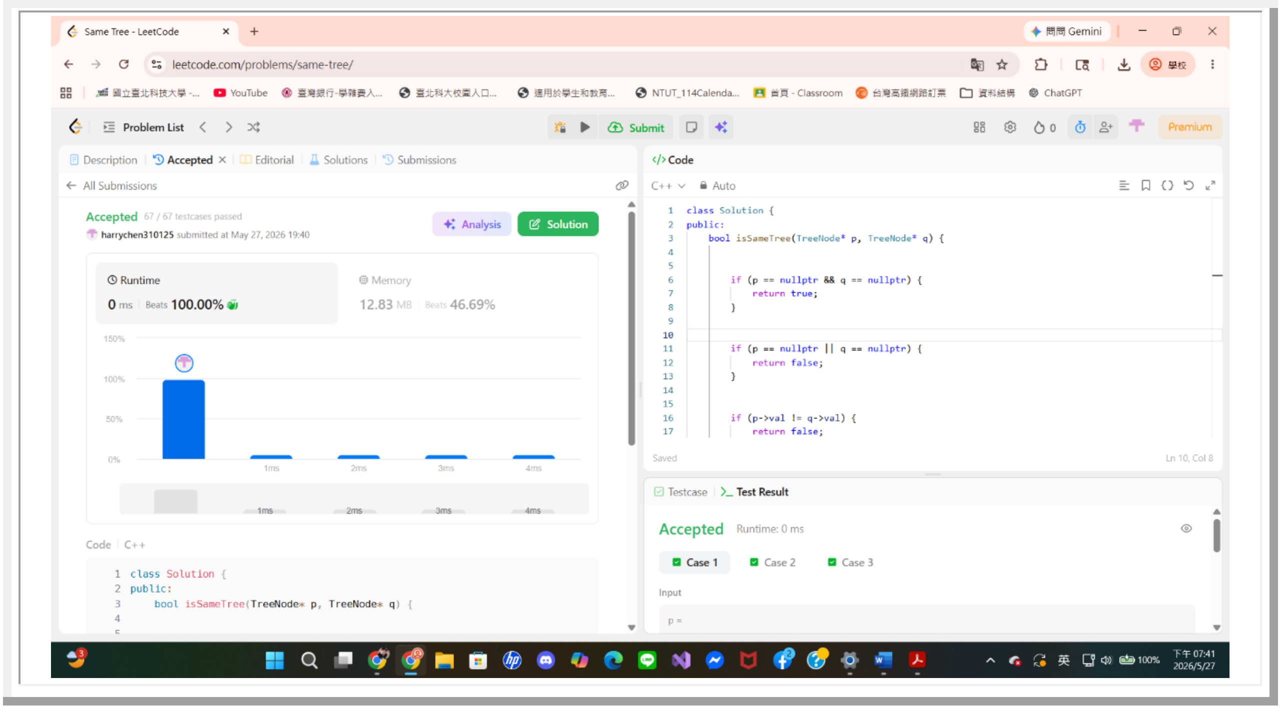
姓名：陳冠豪

學號：114310125

加分題-Q1(Same Tree)
Given the roots of two binary trees p and q, write a function to check if they are the same or not.Two binary trees are considered the same if they are structurally identical, and the nodes have the same value.
Code

```
class Solution {
public:
    bool isSameTree(TreeNode* p, TreeNode* q) {


        //如果兩個節點都沒有資料代表目前這個位置是一樣的
        if (p == nullptr && q == nullptr) {
            return true;
        }


        //如果只有其中一邊是空的代表樹的結構不一樣
        if (p == nullptr || q == nullptr) {
            return false;
        }


        //如果兩個節點的值不同代表不是同一棵樹
        if (p->val != q->val) {
            return false;
        }

        //繼續比較左子樹和右子樹
        //左右兩邊都相同才會回傳 true
        return isSameTree(p->left, q->left) &&
               isSameTree(p->right, q->right);
    }
};
```

Result


Same Tree - LeetCode
leetcode.com/problems/same-tree/
Problem List
Submit
Premium
Description
Accepted
Editorial
Solutions
Submissions
All Submissions
Accepted 67 / 67 testcases passed
harrychen310125 submitted at May 27, 2026 19:40
Analysis
Solution
Runtime
0 ms Beats 100.00%
Memory
12.83 MB Beats 46.69%
Code C++
class Solution {
public:
bool isSameTree(TreeNode* p, TreeNode* q) {
if (p == nullptr && q == nullptr) {
return true;
}
if (p == nullptr || q == nullptr) {
return false;
}
if (p->val != q->val) {
return false;
Testcase
Test Result
Accepted Runtime: 0 ms
Case 1
Case 2
Case 3
Input
2026/5/27